

Final Assessment Study Guide

BTP610 -- Mobile Applications

To prepare for the final assessment, focus on the following key areas:

1. Firebase Setup & Configuration

Project Setup

- Installing and importing the firebase package
- `initializeApp()`, `getAuth()`, and `getFirestore()` -- what each one connects to and returns
- `firebaseConfig.ts` -- what it holds and why it should never be shared publicly
- Exporting shared instances (auth, database) for use across screens

2. Firebase Authentication

Creating & Verifying Accounts

- Creating a new account vs. signing an existing user in -- which function does which
- What gets returned after a successful sign up or sign in
- Reading a signed-in user's information (e.g. their unique ID, their email)

Auth State

- Listening for changes in login status as they happen
- The different states a "who is logged in" value can be in, and what each one means for the app
- Signing a user out and what happens to the rest of the app afterward
- Error handling around authentication calls

3. Firestore Data Model & CRUD

Data Model

- Collections vs. documents -- how data is organized
- Auto-generated document IDs vs. IDs you choose yourself, and when to use each
- Linking a Firestore document to a specific user

Create / Read / Update / Delete

- Adding a new document to a collection
- Writing to a document at a specific, known ID
- Reading a document's data one time vs. listening for ongoing changes
- Updating only specific fields on an existing document without touching the rest
- Permanently removing a document
- Building a reference to a collection or a specific document

Real-Time Listeners

- Setting up a listener that keeps the on-screen list in sync with the database
- Turning the data a listener sends back into a usable array
- Why listeners need to be cleaned up, and when that cleanup happens

4. React Hooks & State Patterns

- `useState` and `useEffect` -- core behavior and common uses

- Dependency arrays -- what an empty array means vs. leaving it off entirely
- useRef -- when to reach for a ref instead of state
- Cleanup functions returned from useEffect
- Storing an entire form as one object in state, and updating a single field without losing the others
- Writing a custom hook that wraps a listener and returns a value to any screen that calls it

5. Navigation & Routing (Expo Router)

File-Based Routing

- How file and folder names map to screens and routes
- Grouping related screens into a folder without changing the URL structure
- Deciding where to send the user based on their login status

Stacks & Tabs

- Hiding or customizing headers for a group of screens
- Setting up a tab bar with multiple screens, icons, and an active color
- Adding a button or icon to a screen's header (e.g. for signing out)
- Navigating in a way that lets the user go back vs. replacing the screen so they can't

6. Lists & Components

- Efficiently rendering a list of items from state or Firestore
- Giving each item in a list a unique key
- Defining a small reusable component for one item in a list
- Displaying different pieces of data with different styling within an item

7. Camera & Media Library

Permissions

- Requesting and checking camera permission
- Requesting and checking media library (photo-saving) permission separately
- Only rendering the camera once permission has actually been confirmed

Capturing & Saving Photos

- Referencing the live camera view so it can be controlled from code
- Switching between front and back camera
- Capturing a photo from the live preview
- Saving a captured photo into the device's photo gallery
- Safely guarding any camera action that depends on the camera already being mounted

8. Forms & User Input

- Connecting a text input's value and change handler to state
- Validating required fields before submitting
- Showing and hiding an error message based on state
- Useful input props: hiding password characters, controlling keyboard type, controlling auto-capitalization

9. Application Flow

You should be comfortable tracing how a full feature works end to end, for example:

- A user creates an account, and their profile is saved separately from their login credentials
- A user signs in, and the app automatically routes them based on their login state
- A user adds, edits, or deletes something, and the change appears on screen without a manual refresh
- A user signs out, and the app automatically returns them to the login screen

Focus on how the pieces connect -- what triggers what -- not just what an individual line of code does in isolation.

10. Important Notes

- Written in TypeScript, using Expo and React Native
- Firebase Authentication handles login; Firestore handles all stored data
- Follow the coding patterns used in class: typed hooks, object state with spread updates, try/catch around async calls
- Review the “Common Mistakes” callouts included in the later slide decks -- they point directly at the kinds of errors worth knowing how to spot and fix

11. Key Advice

- Rebuild small pieces of the project from memory instead of re-reading the slides passively
- Practice reading a snippet of given code and predicting exactly what it will do, step by step
- Practice writing a small function (an auth call, a Firestore read/write, a permission check) without looking anything up
- Make sure you understand *why* each pattern is used, not just how to type it out
- Use the practice review test alongside this guide for hands-on drilling